Tianjiao Sun

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EDUCATION

Northwestern University *Chicago, IL, USA* *September 2014 - December 2020*

Feinberg School of Medicine

Dual Degree Program:

- Degree: Doctorate （December 2020）
- Major Concentration: Life Sciences
- Thesis advisor: Dr. Douglas E. Vaughan, MD

- Degree: Master of Science in Clinical Investigation (December 2020）
- Major Concentration: Clinical Investigation

Kellogg School of Management

Certificate Program: Management for Scientists and Engineers *June 2019 – August 2019*

- Courses: Marketing, Accounting for Decision Making, Economics & Strategy, Business Decisions under Uncertainty, Crisis Management, Negotiations, Leadership & Teams, Finance, Entrepreneurship & Innovation, Marketing Analytics, Operations Management, IP Management.

Illinois Institute of Technology *Chicago, IL, USA* *August 2010 - May 2012*

- Degree: Master of Science in Biology
- Major Concentration: Cell and Molecular Biology

Shanghai University of Traditional Chinese Medicine *Shanghai, China* *September 2002- July 2007*

- Degree: Bachelor of Medicine
- Major Concentration: Clinical Medicine (Including both Western Medicine and Traditional Chinese Medicine)
- Graduated with Honor

CLINICAL EXPERIENCE

Resident Physician in Internal Medicine *July 2007 to August 2010*

Longhua Hospital Affiliated to Shanghai University of Traditional Chinese Medicine *Shanghai, China*

- Developed diagnosis and formulated therapeutic strategies for about 1,000 patients in Divisions of Neurology, Endocrinology, Cardiology, Nephrology, Gastroenterology
- Performed routine operations of internal medicine

ACADEMIC RESEARCH EXPERIENCE

PhD Student *June 2015 to December 2020*

Northwestern University / Dr. Douglas Vaughan's Lab *Chicago, IL, USA*

Thesis Project: Role of Plasminogen Activator System in Endothelial Senescence and Cardiovascular Aging

1) The role of Plasminogen Activator Inhibitor 1(PAI-1) in homocysteine-induced vascular aging
- Investigated the role of PAI-1 in homocysteine-induced endothelial senescence *in vitro*
- Investigated the role of PAI-1 in homocysteine-induced cardiovascular aging in vivo

2) The role of PAI-1 in Chronic Kidney Diseases (CKD) and CKD-induced cardiovascular diseases (CVDs)
- Established the chronic kidney disease mouse models in the lab and investigated the role of PAI-1 in these models

Other projects: The role of Acetyltransferase p300 in Hypertension-induced Cardiac Hypertrophy and Fibrosis

- Tested the efficacy of acetyltransferase p300 inhibitor, L002 in reversing hypertension-induced cardiac hypertrophy and fibrosis in Angiotensin II-induced hypertension model

Research Assistant *June 2013 to August 2014*

Northwestern University / Dr. Gangjian Qin's Lab *Chicago, IL, USA*
Project: Molecular Mechanisms Regulating Stem Cells in Ischemic Cardiac Repair

1) The crosstalk of MCP-1/CCR2 and SDF-1/CXCR4 pathways on stem cell homing to ischemic tissue
   - Investigated the effect of CCR2 antagonism on post-MI recovery by sensitizing stem cell SDF-1/CXCR4 pathway and homing.
2) The role of E2F1 in the regulation of bone marrow progenitor cell oxidative metabolism and ischemic cardiac repair
   - Analyzed the differentiation of bone marrow stem cells from both E2F1 knockout and wild-type mice toward Endothelial Progenitor Cells (EPC) (using endothelial markers VE-Cadherin, vWF, Flk1, CD31) under hypoxic and normoxic conditions
   - Detected the impact of E2F1 gene knockout in cell metabolism by analyzing pyruvate dehydrogenase kinase (isozyme1,2,3,4) under hypoxic and normoxic conditions
   - Detected the impact of E2F1 gene knockout in cell apoptosis under hypoxic and normoxic conditions

Research Assistant *September 2011 to May 2013*
University of Chicago / Dr. Fotini Gounari's lab *Chicago, IL, USA*
Project: Molecular Mechanisms Governing Lymphocyte Development and Transformation

1) Role of *Wnt* Pathway in regulating the development of CD4 T cells
   - Investigated periphery T-cell activation using two conditional mouse models (CD4CrecAPC YFP and CD4CreER CatYFP)
   - Standardized Bisulfite Sequencing Method for detecting demethylation pattern of FoxP3 Locus
2) Role of *Wnt* Pathway in the maintenance of genomic stability in thymocytes
   - Analyzed the sequence composition of coding joins (CJ) to access possible defects in DNA repair in wild-type and β -catenin activated mouse model

Research Assistant *July 2011 to August 2011*
University of Illinois at Chicago/Dr. Viswanathan Nataraja's Lab *Chicago, IL, USA*
Project: The Role of Sphingosine-1-phosphate (S1P) in Acute and Chronic Lung Injury

- Analyzed protein and mRNA expression of fibrosis markers (including alpha-smooth muscle actin and fibronectin), S1P receptors (EDG1, 3, 5, 6, 8), and TGF-beta in S1P1 knockout and wild-type mice and in human lung fibroblasts; and investigated their contribution to the pathogenesis of pulmonary fibrosis

OTHER RELATED EXPERIENCE

Kids-Inspired Innovation for Careers in Science (KIICS) Scholar *Oct 2019 to May 2020*
Ann & Robert H. Lurie Children's Hospital *Chicago, IL, USA*

- Shadowed physicians on rounds at Lurie Children's
- Participated in clinical journal clubs and case study conferences
- Shared observations made during the clinical experience and discussed the case and the science behind the case with the KIICS group

TEACHING EXPERIENCE

Student Mentor, Student Assisted Mentoring Program (StAMP)*, Chicago, IL* *September 2019 to August 2020*
Northwestern University *Chicago, IL, USA*

- Mentored a first-year PhD student in course, qualifying exam, lab rotation and other graduate student issues

Teaching Assistant *March 2017 to May 2017*
Northwestern University *Chicago, IL, USA*
Course: Receptors and Signaling Mechanisms

- Facilitated the course Receptors and Signaling Mechanisms by promoting the in-class discussion, organizing the class materials, and timely and efficient communication with both the professors and the students

Student Mentor *June 2013 to August 2014*

Northwestern University/ Dr. Gangjian Qin's Lab *Chicago, IL, USA*

- Mentored high school students in performing experiments, writing research summaries, and presenting ideas and data

Student Mentor and Instructor *July 2007 to August 2010*
Longhua Hospital Affiliated to Shanghai University of Traditional Chinese Medicine *Shanghai, China*

- Mentored 60 medical students during their internship, including training them in clinical operation, improving their doctor-patient communication skills and expanding their knowledge in clinical medicine
- Lectured in diagnosis and treatment of common diseases in internal medicine and organized discussions on specific cases among students

Course Tutor for International Students *July 2003 to August 2005*
Shanghai University of Traditional Chinese Medicine *Shanghai, China*

LEADERSHIP

Event Chair, Women in Science and Engineering Research (WISER) *June 2018-June 2020*
Northwestern University

- Organized the Chicago campus events including inviting women faculty seminar speakers, managing the logistics and advertising the events

Co-Founder and Coordinator, NU Circles for Women in STEM *June 2018-June 2019*
Northwestern University

- Initiated the small peer groups to have open conversations about the various challenges faced in academia by women by collaborating with students from Interdepartmental Neuroscience Program and securing the funding support
- Managed the regular meetings by planning the agenda and promoting the events

Coordinator, Northwestern Chinese Association (NWCA) *September 2014- June 2015*
Northwestern University (Chicago Campus) *Chicago, IL, USA*

- Co-founded and coordinated the English Corner for visiting scholars from China, and helped the visiting scholars to adjust to the new language and cultural environment

Coordinator, Department of Human Resources *May 2010- August 2010*
Longhua Hospital Affiliated to Shanghai University of Traditional Chinese Medicine *Shanghai, China*

Editor-in-chief, Journal of Longhua Clinical Medical School *September 2004- July 2005*
Shanghai University of Traditional Chinese Medicine *Shanghai, China*

VOLUNTEER AND OTHER ACTIVITIES

Volunteer, Cancer Survivors' Celebration Walk & 5K *June 2016, June 2018, June 2019*
Robert H. Lurie Comprehensive Cancer Center (Northwestern University) *Chicago, IL, USA*

Volunteer, Science Fest *May 2018*
Illinois Science Council *Chicago, IL, USA*

Volunteer and Organizer, Visiting the elderly in the Social Welfare Institute in Pudong New District *2005*
Shanghai University of Traditional Chinese Medicine *Shanghai, China*

AWARDS AND SCHOLARSHIPS

American Heart Association Predoctoral Fellowship *2018*
College Graduate Excellence Award of Shanghai *2007*
Excellent Student Cadre of Shanghai University of Traditional Chinese Medicine *2006*
The Second Prize Scholarship of Shanghai University of Traditional Chinese Medicine *2006, 2004, 2003*
The First Prize Scholarship of Shanghai University of Traditional Chinese Medicine *2005*
Outstanding Student Award of Shanghai *2005*

PUBLICATIONS

Research Papers

Sun T, Ghosh AK, Eren M, Lux E, Miyata T, Vaughan DE. PAI-1 contributes to Homocysteine-induced Cellular Senescence. *Cellular signaling. 2019 Dec 1;64:109394.*

Rai R, **Sun T**, Ramirez V, Lux E, Eren M, Vaughan DE, Ghosh AK. Acetyltransferase p300 inhibitor reverses hypertension - induced cardiac fibrosis. *Journal of cellular and molecular medicine. 2019 Apr;23(4):3026-31.*

Xu S, Tao J, Yang L, Zhang E, Boriboun C, Zhou J, **Sun T**, Cheng M, Huang K, Shi J, Dong N, Liu Q, Zhao TC, Qiu H, Harris RA, Chandel NS, Losordo DW, Qin G. E2F1 suppresses oxidative metabolism and endothelial differentiation of bone marrow progenitor cells. *Circulation research. 2018 Jan 22:CIRCRESAHA-117.*

Cao K, Collings CK, Marshall SA, Morgan MA, Rendleman EJ, Wang L, Sze CC, **Sun T**, Bartom ET, Shilatifard A. SET1A/COMPASS and shadow enhancers in the regulation of homeotic gene expression. *Genes & development. 2017 Apr 15;31(8):787-801.*

Keerthivasan S, Aghajani K, Dose M, Molinero L, Khan MW, Venkateswaran V, Weber C, Emmanuel AO, **Sun T**, Bentrem DJ, Mulcahy M, Keshavarzian A, Ramos EM, Blatner N, Khazaie K, Gounari F. β-Catenin promotes colitis and colon cancer through imprinting of proinflammatory properties in T cells. *Science translational medicine. 2014 Feb 26;6(225):225ra28.*

Dose M, Emmanuel AO, Chaumeil J, Zhang J, **Sun T**, Germar K, Aghajani K, Davis EM, Keerthivasan S, Bredemeyer AL, Sleckman BP, Rosen ST, Skok JA, Le Beau MM, Georgopoulos K, Gounari F. β-Catenin induces T-cell transformation by promoting genomic instability. *Proceedings of the National Academy of Sciences. 2014 Jan 7;111(1):391-6.*

Huang LS, Fu P, Patel P, Harijith A, **Sun T**, Zhao Y, Garcia JG, Chun J, Natarajan V. Deficiency Confers Protection against Bleomycin-Induced Lung Injury and Fibrosis in Mice. *American journal of respiratory cell and molecular biology. 2013 Dec;49(6):912-22.*

Book Chapter

Chen FX, Marshall SA, Deng Y, **Sun T**. (2018). Measuring Nascent Transcripts by Nascent-seq. *Methods in Molecular Biology. 2018;1712:19-26.*

CONFERENCE ABSTRACTS AND POSTERS

Sun T, Rai R, Ramirez V, Lux E, Eren M, Vaughan D, Ghosh A. Small Molecule Inhibitors of p300 Acetyltransferase Activity Reverse Hypertension-induced Cardiac Fibrosis. *CIRCULATION RESEARCH 2018 Dec 7 (Vol. 123, No. 12, pp. E73-E73).*

Sun T, Rai R, Ramirez V, Lux E, Vaughan DE, Ghosh AK. Acetyltransferase p300 inhibitor, L002 reverses hypertension-induced cardiac hypertrophy and fibrosis.
American Society for Matrix Biology Meeting 2018

Sun T, Ghosh AK, Ramirez V, Lux E, Miyata T, Vaughan DE. Inhibition of plasminogen activator inhibitor-1(PAI-1) attenuates homocysteine-induced endothelial senescence and cardiovascular pathology.
i) 14th Annual Lewis Landsberg Research Day 2018
ii) Feinberg Cardiovascular Research Day 2018

Levine JA, **Sun T**, Eren M, Lux E, Miyata T, Vaughan DE. Reversal of diet induced metabolic syndrome in mice with an orally active small molecule inhibitor of PAI-1.
ENDO 2019

Xu S, **Sun T**, Tao J, Qin G. Deletion of E2f1 Promotes Bone-marrow Progenitor Cell Differentiation and Ischemic Cardiac Repair. *Circulation. 2014;130:A18478*